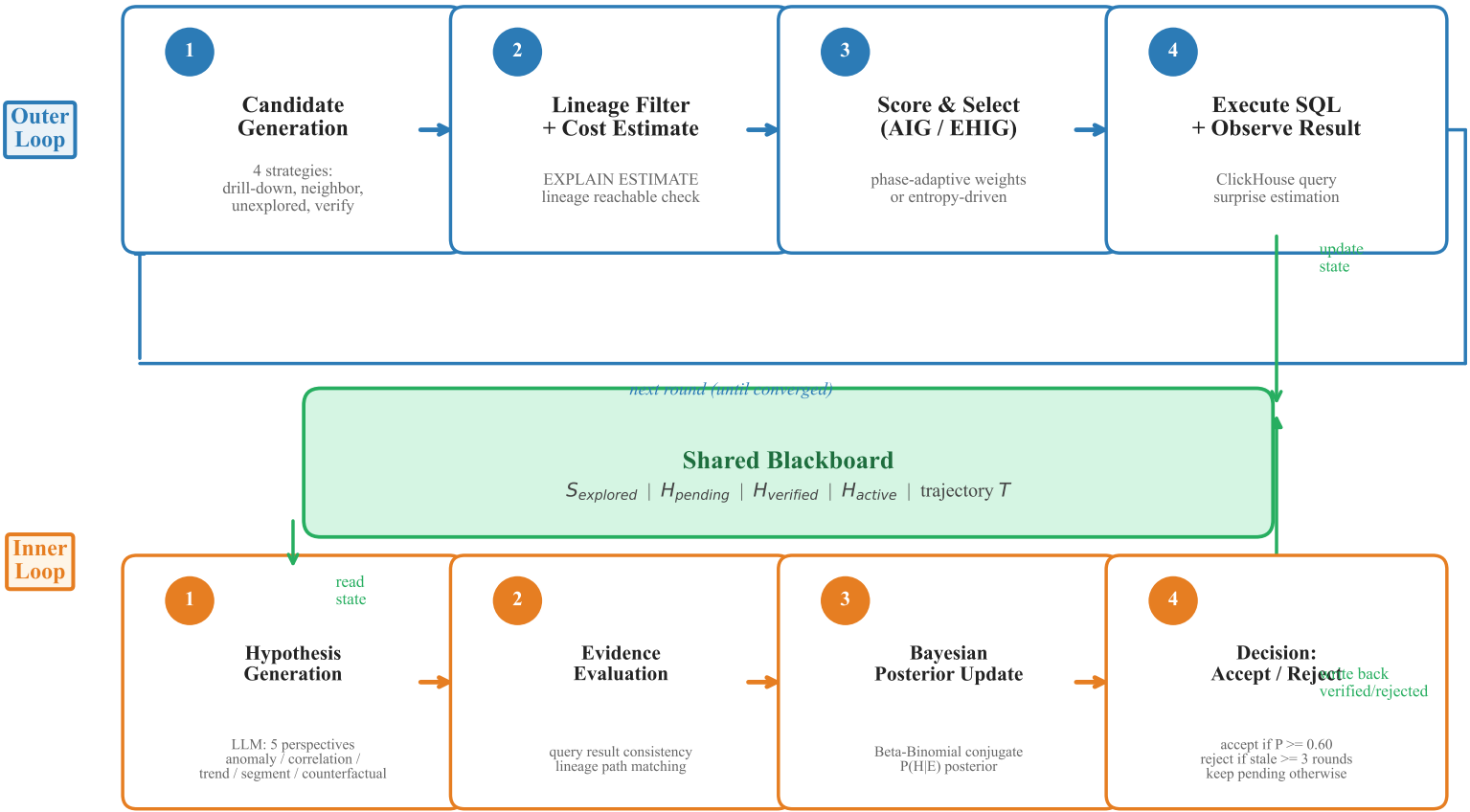

Outer Loop
1
Candidate Generation
4 strategies: drill-down, neighbor, unexplored, verify
2
Lineage Filter + Cost Estimate
EXPLAIN ESTIMATE lineage reachable check
3
Score & Select (AIG / EHIG)
phase-adaptive weights or entropy-driven
4
Execute SQL + Observe Result
ClickHouse query surprise estimation
update state
next round (until converged)
Shared Blackboard
$S_{explored}$ | $H_{pending}$ | $H_{verified}$ | H_{active} | trajectory T
Inner Loop
read state
1
Hypothesis Generation
LLM: 5 perspectives anomaly / correlation / trend / segment / counterfactual
2
Evidence Evaluation
query result consistency lineage path matching
3
Bayesian Posterior Update
Beta-Binomial conjugate P(H|E) posterior
4
Decision: Accept / Reject
write back verified/rejected
accept if P >= 0.60
reject if stale >= 3 rounds
keep pending otherwise